Maximizing Solar Panel Output for a Fixed Area

In this project, you will use MATLAB to formulate and solve an optimization problem: Given a fixed area, determine the optimal tilt angle and aspect ratio of a solar panel to maximize the total energy output.

Use this Live Script as a template to get started on your project solution. You may include helper functions at the bottom of this document or as separate files. For a tutorial on how to use Live Scripts, check out this video.

Suggested Tasks

1. Define the objective function in MATLAB (the energy function).
2. Use a problem-based optimization workflow to find the values of θ and r that maximize the energy output. Constrain the values: $0 \le \theta \le 90$ and $0.5 \le r \le 4$
3. Plot the objective function using `meshgrid` and `surf` to visualize E(θ, r) as a 3D surface plot. Then, print the optimal angle, ratio, and corresponding energy output.

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Break Down the Problem

In the space below, define the problem statement:

List the requirements, constraints, and success criteria for the project:

What is your proposed solution?

List the steps you will need to take to achieve your proposed solution:

What challenges do you face in the process of achieving the solution?

Task 1: Define the Objective Function

The energy function is the objective function for this optimization problem. It calculates how much energy a solar panel produces based on its tilt angle (theta) and aspect ratio (r). Check the Project Description for the energy formula and parameter definitions and use the image below as a guide:

$A = 2m^2$

$sunIntensity(\theta) = 1000 * \cos(\theta - 45)$

$$E(\theta, r) = A \cdot \eta(\theta) \cdot sunIntensity(\theta) \cdot f(r)$$

$f(r) = \exp(-0.1 * (r-1)^2)$

$\eta(\theta) = \cos(\theta - 30)$

```
% Insert your code in the spaces provided (or make use of helper functions
and indicate where
% they can be found). Ensure your code is well-documented.

% Your energy function should output energy units and take as inputs tilt
% angle (theta) and aspect ratio (r). Using the intermediary steps can help
% simplify how you define E.

% Step 1: Define A (panel area in m^2)

% Step 2: Define nu (efficiency function based on tilt angle)

% Step 3: Define sunIntensity (sunlight variation with tilt)

% Step 4: Define fr (shape efficiency factor based on aspect ratio)

% Step 5: Calculate E (total energy output)
```

Task 2: Implement the Optimization Function

Use MATLAB's Optimization Toolbox to find the optimal tilt angle and aspect ratio that MAXIMIZE energy output. This MATLAB doc page will be helpful.

```
% Insert your code in the spaces provided (or make use of helper functions
and indicate where
% they can be found). Ensure your code is well-documented. The intermediary
% steps below take you through a problem-based optimization workflow.

% Step 1. Create the optimization problem. We're telling MATLAB: "We want
to MAXIMIZE something."
% That "something" will be energy.

% Step 2. Define the optimization variables we can change in the problem
% These are the values MATLAB will try to optimize:
% - theta (tilt angle), must stay between 0 and 90 degrees
% - r (aspect ratio), must stay between 0.5 and 4
```

```
% Step 3. Turn our energy function into an optimization problem MATLAB can
work
% with.
% This step wraps our energy function so MATLAB can plug in different
% theta and r values while solving the problem.
% note: we're optimizing our energy function, so we should try to convert
% that function into an optimization expression

% Step 4. Tell MATLAB what we want to maximize
% We set our energy formula as the "objective" of the problem
% (this is what we want the solver to maximize)

% Step 5. Give MATLAB a starting point to begin the search
% Optimization needs a guess to start from:
    % Start searching from a tilt angle of ? degrees
    % Start searching from an aspect ratio of ?

% Step 6. Now solve the problem!
% Hint: look for a Optimization Toolbox function meant to solve
% expressions. Store the outputs for plotting purposes later on.
```

Task 3: Output the Results

Use the appropriate plotting function to plot E(θ, r) and visualize the energy surface. Print the optimal tilt angle and aspect ratio, and corresponding energy output.

```
% Insert your code in the spaces provided (or make use of helper functions
and indicate where
% they can be found). Ensure your code is well-documented. The intermediary
% steps below will guide you through creating a 3D surface plot.

% Step 1: Make a bunch of points for the x and y axis meshgrid
% The step for the angle is 1, the step for ratio is 0.1
% Hint: keep theta and r within the constraints specified in the problem

% Step 2: Calculate energy for every point on the meshgrid using element-
wise operations

% Step 3: Draw the 3d graph using the "surf" function

% Step 4: Label everything on the surf plot (title, axes, etc)

% Step 5: Print the optimal tilt angle and aspect ratio, and corresponding
% energy output to the command window
```

Interpretation of Results

Summarize your findings by interpreting their physical/engineering meaning:

What are some limitations of your work?

What are practical next steps?